

Detecting industrial trawler incursions into artisanal fishing zones along the Tamil Nadu and Puducherry coast: a Sentinel-2 and AIS framework for evidence-based small-scale fisheries governance

Project Lead: [HOD FULL NAME], Head of Department, Artificial Intelligence & Machine Learning, [UNIVERSITY NAME]

Co-Investigators: [YOUR FULL NAME] (Lead Developer); [CLASSMATE FULL NAME]

Location of applicant: [UNIVERSITY NAME], Chennai / Puducherry, Tamil Nadu, India | **Location of proposed work:** Tamil Nadu and Puducherry coastal waters, India (Bay of Bengal, FAO Area 57)

1. Background and significance

Tamil Nadu and Puducherry's 1,076 km coastline supports more than 600,000 marine fishers, of whom roughly 90% operate non-mechanised and motorised craft within state-regulated artisanal zones. The **Tamil Nadu Marine Fisheries Regulation Act (1983)** reserves the first three nautical miles from shore exclusively for these small-scale fishers, prohibits mechanised trawling within this zone, and imposes an annual June–August fishing ban to protect spawning. In practice, enforcement is sparse: state patrols are concentrated in a handful of harbours, and violent confrontations between artisanal fishers and intruding mechanised trawlers are reported every month across Cuddalore, Nagapattinam, Karaikal, Pudukkottai, Ramanathapuram and Thoothukudi. There is presently **no independent, satellite-based, public monitoring of zonal compliance** anywhere on India's southeast coast.

As Global Fishing Watch's Chief Scientist David Kroodsma noted in the 2026 Research Roadmap, “*determining the integrity of artisanal fishing zones — and whether industrial vessels are staying out of them*” is a foundational small-scale-fishery question that GFW's new Sentinel-2 optical detection layer is uniquely positioned to answer. This proposal addresses that question directly for one of the Indian Ocean's most populous, most contested, and most data-poor small-scale fishery regions.

2. Research questions

Using publicly available GFW data, the project will quantify, for 2022–2025: **(i)** the frequency, spatial distribution and seasonality of mechanised vessel incursions inside the legally protected artisanal zone (≤ 3 nm) along the Tamil Nadu and Puducherry coast; **(ii)** the degree of compliance with the annual June–August mechanised-fishing ban, by district; and **(iii)** the change in industrial pressure on artisanal grounds over the period, including seasonal, monsoonal and oceanographic drivers.

3. Methods and GFW datasets used

Sentinel-2 optical vessel detections (primary). GFW's 2025 Sentinel-2 release (10 m resolution, near-shore capable) is the single most appropriate dataset for the wooden- and FRP-hulled mechanised trawlers (12–18 m) that dominate the Tamil Nadu fleet and are largely invisible to AIS and to coarser SAR pipelines. We will retrieve all Sentinel-2 vessel detections within a 0–12 nm coastal buffer along the entire Tamil Nadu and Puducherry coast for 2022–2025 via the GFW API.

Sentinel-1 SAR vessel detections (complementary). GFW's SAR layer (Paolo et al., *Nature* 2024) provides all-weather, day-and-night detections that complement Sentinel-2 during the southwest and northeast monsoon cloud-cover periods and serve as an independent cross-check on optical detection counts.

AIS apparent fishing effort (context). Although most of the relevant fleet does not broadcast AIS, the GFW AIS layer characterises the larger industrial component — multi-day trawlers >24 m and foreign distant-water vessels in the wider Bay of Bengal — and identifies AIS-broadcasting violators where they exist.

Pipeline. Detections will be (i) spatially classified relative to a legal-zone polygon constructed from notifications under the TNMFR Act; (ii) clustered using density-based and trajectory-based methods (HDBSCAN, ST-DBSCAN) to distinguish single transits from sustained fishing; and (iii) modelled as a function of district, season and oceanographic covariates (sea-surface temperature, chlorophyll-a, monsoon indices) using gradient-boosted regression. A LightGBM classifier will assign each detection a *likely-fishing-in-zone* probability. The full pipeline will be released as an open-source Python package built on the official `gfw-api-python-client`, with reproducible Jupyter notebooks and a public dashboard.

Validation. Outputs will be cross-validated against (i) Tamil Nadu Coastal Security Group and state fisheries patrol incident records, (ii) harbour-landing records and Marine Fisheries Census 2016/2025 village-level vessel counts published by CMFRI, and (iii) where possible, geo-tagged community reports collected through fisher-association partners.

4. Alignment with Global Fishing Watch priorities

The project advances three of GFW's stated impact areas: **support for small-scale fishers, IUU fishing detection and deterrence**, and **evidence-based fisheries management**. It directly operationalises the 2026 Research Roadmap's flagship small-scale-fleet objective in a representative shallow tropical environment, and contributes a methodology paper of broad utility to coastal-state users of the GFW platform across the Indian Ocean rim — an area the GFW-FAO Atlas review identifies as a known coverage gap.

5. Team, feasibility and data-availability check

[HOD FULL NAME] heads [UNIVERSITY NAME]'s Department of Artificial Intelligence and Machine Learning, with [N] years of research and teaching experience in [DOMAINS] and prior projects on [RELEVANT AREAS]. [YOUR FULL NAME] is a second-year B.Tech (AI/ML) student and the project's lead developer, with demonstrated technical experience in [GitHub portfolio / hackathons / prior ML projects – cite 1–2 specifics]. [CLASSMATE FULL NAME] supports data engineering. The team is based in Chennai and Puducherry, immediately adjacent to the study area, fluent in Tamil, and able to engage local fishing communities and state fisheries officials directly.

Data availability. We have verified GFW data availability for the Tamil Nadu and Puducherry coastal bounding box (approx. 8°–14°N, 77°–80.5°E) on the Global Fishing Watch data and tool catalog. AIS apparent fishing effort, Sentinel-1 SAR vessel detections, and the new Sentinel-2 optical vessel detections all return populated results across the 2022–2025 study window. A preliminary detection density figure generated by the lead developer is available on request.

6. Timeline and deliverables

The project runs 18 months (December 2026 – May 2028). **Months 1–4:** data ingestion, artisanal-zone polygon construction, Sentinel-2 retrieval and QC. **Months 5–9:** clustering, classifier training, validation. **Months 10–13:** district-level compliance analysis and oceanographic modelling. **Months 14–16:** manuscript preparation (target: *Marine Policy* or *Ocean & Coastal Management*). **Months 17–18:** open dataset release, public dashboard launch, dissemination. Deliverables: a peer-reviewed manuscript; an open-access processed dataset on Zenodo; an MIT-licensed Python package and reproducible notebooks on GitHub; an interactive public dashboard; and a 2-page policy brief in English and Tamil.

7. Budget summary

Total request: **USD 20,000 (Group)**. Indicative allocation: Lead Developer stipend (18 mo) USD 8,000; Project Lead supervision USD 4,000; Co-Investigator stipend USD 1,500; cloud compute (GFW API, Earth Engine, model training) USD 3,000; data download and storage USD 800; open-access publication fee USD 1,500; dissemination (Tamil-language brief, dashboard hosting, workshop) USD 1,200. **No indirect or institutional overhead costs are requested**, in accordance with GFW guidelines.